Yinuo Han

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RESEARCH INTERESTS

- Exoplanetary systems, debris disks, planet formation, disk substructure, dynamical evolution
- High-angular-resolution imaging, high-contrast optical/infrared imaging, millimeter interferometry
- Dust formation, carbon feedback, Wolf-Rayet stars, colliding-wind binaries

ACADEMIC APPOINTMENTS

1. **Melza M. and Frank Theodore Barr Fellow, Caltech, USA** Since Oct 2024
   Postdoctoral fellowship, Planetary Sciences, Division of Geological and Planetary Sciences

EDUCATION

1. **PhD in Astronomy, University of Cambridge, UK** Oct 2020 - Jul 2024
   Institute of Astronomy · Trinity College · Gates Cambridge Scholar
   Thesis: *The Structure of Extrasolar Planetesimal Belts in Images*
   Advisor: Prof. Mark Wyatt
2. **BSc (Adv) (Hons), University of Sydney, Australia** Feb 2016 - Dec 2019
   Majors in Physics, Neuroscience · First Class Honours (Physics) and the University Medal
   Thesis: *A Dance with Dragons: The Enigmatic Wolf-Rayet Binary Apep*
   Advisor: Prof. Peter Tuthill

   Study Abroad, Physics, University of California, Berkeley, USA Aug 2017 - Dec 2017

ADVISING

List of theses and research projects advised.

- *VLT observations of β Pic and its dust clump.* Since Jun 2025
  Summer research project/Caltech SURF by Polaris Hayes (Caltech).
- *ALMA imaging of WR112 and dust modeling with JWST/MIRI.* Since Jun 2025
  Summer research project/Caltech SURF by Donglin Wu (Yale University).
- *Joint ALMA and JWST/NIRCam modeling of the vertical structure of the AU Mic debris disk.*
  Summer research project/Caltech SURF by Junyi Zhao (University of Michigan). Since Jun 2025
- *Secular perturbation as the origin of the dust and gas clump of Beta Pictoris.* Oct 2022 – Jun 2023
  Cambridge Part III/Masters project by Faran Sarwar (co-advised with Mark Wyatt).
- *The structure of the Beta Pictoris debris disk inferred from multi-wavelength mid-infrared images.*
  Summer research project by Andrew Zhang. Jul – Aug 2022

TEACHING

1. **Topics in Astrophysics** (Distributions, Timescales, Tides, Planet Formation) 2023
   Institute of Astronomy, University of Cambridge
   Regular supervisions/tutorial classes for Part II Astrophysics/final-year undergraduate astrophysics.
2. **Structure and Evolution of Stars** 2022
   Institute of Astronomy, University of Cambridge
   Regular supervisions/tutorial classes for Part II Astrophysics/final-year undergraduate astrophysics.
3. **Oscillating Systems, Waves and Quantum Waves, Fields** 2021
   Trinity College, University of Cambridge
   Weekly supervisions/tutorial classes for Part IA Physics/first-year undergraduate physics.
4. **Physics 1B** (Electromagnetism, Fluids, Quantum Mechanics) 2019
   School of Physics, University of Sydney
   Weekly tutorial classes for first-year undergraduate physics.

OBSERVING

- *ALMA co-I. program*: Testing a new picture of debris disk gas by observing CI towards HD 121617 (ID 2025.1.00273.S, P.I. Gianni Cataldi) C.12, 2025
- *VLT/VISIR P.I. program*: Testing the giant impact scenario to explain the dust clump and cat's tail in Beta Pictoris (114.27CQ) P114, 2024
- *ALMA P.I. program*: First resolved ALMA imaging of carbon-rich Wolf-Rayet binary dust structures (2024.1.00803.S) C.11, 2024
- *ALMA co-I. program*: Resolving the extended structure associated with Apep's enigmatic wind (2024.1.01244.S, P.I. Joseph Callingham) C.10, 2024
- *JWST P.I. program*: What lies beyond the inner spiral of Apep? (GO 5842) C.3, 2024
- *JWST P.I. program*: What causes warm dust interior to planetesimal belts? (GO 5709) C.3, 2024
- *VLT/VISIR P.I. program*: Wind kinematics in the extreme colliding-wind binary Apep: the first Galactic long-duration gamma-ray burst progenitor? (113.26A7) P113, 2024
- *ALMA P.I. program*: First ALMA imaging of dust around Wolf-Rayet binaries (2023.1.00999.S) C.10, 2023
- *ALMA co-I. program*: Vertical structure and planetary system dynamics (2023.1.00487.S and 2025.1.00362.S, P.I. Meredith Hughes) C.10/12, 2023/2025
- *JWST co-I. program*: Fingerprinting the history of episodic dust creation in Wolf-Rayet binaries (4093, P.I. Noel Richardson) C.2, 2023
- *Australia Telescope Compact Array*: Executing observations of Wolf-Rayet binary Apep 2019

PRESS

- *NRAO press release based on Donglin Wu's summer research published in Wu et al. 2026.* Feb 2026
  Yale press release.
- *Caltech press release based on Han et al. 2026.* Jan 2026
  Other ARKS collaboration press releases from NRAO, ALMA and ERC.
- *NASA, STScI and Caltech press releases based on Han et al. 2025.* Nov 2025
  Featured on Astronomy Picture of the Day.
- *University of Cambridge press release based on Han et al. 2022.* Oct 2022
  Reported by major science and mainstream news outlets (e.g., ABC, BBC, CNN, The Guardian).
- *University of Sydney press release based on Han et al. 2020.* Oct 2020
  Reported by 110 news items from 21 countries in 8 languages (e.g., CNN).

ACADEMIC AWARDS

University of Cambridge

Gates Cambridge Scholarship 2020
Cambridge International Scholarship, accepted Gates above instead 2020

University of Sydney

University Medal, top award among graduating class in physics 2019
Henry Chamberlain Russell Prize for Astronomy, for best astronomy Honours thesis 2019
University of Sydney Honours Scholarship 2019
Faculty of Science Olympiad Scholarship 2016 - 2019
Deas-Thomson Scholarship in Physics, for top of class in third-year physics 2018
W.I.B. Smith Prize in Experimental Physics 2018
Dean's List of Academic Excellence 2016, 2017, 2018
Sydney Scholars Award 2016 - 2018
Denison Summer Research Scholarship 2016, 2017
Sydney Abroad International Exchange Scholarship, supported second-year physics exchange 2017
University of Sydney Academic Merit Prize 2016
Levey Scholarship in Physics, for top of class in first-year physics 2016
Smith Prize in Experimental Physics 2016

Other

University of Oxford Clarendon Scholarship, declined 2020
International Physics Olympiad Bronze Medal 2015

RESEARCH GRANTS

External Grants

P.I., JWST Program 5842 (USD 78k) 2024 – 2027
P.I., JWST Program 5709 (USD 74k) 2024 – 2027

Internal Grants

Barr Fellowship research grant, Caltech (USD 30k) 2024 – 2027
Gates Cambridge Scholarship research funding, University of Cambridge (approx. USD 3k) 2020 – 2024
Rouse Ball research funding, Trinity College, Cambridge (approx. USD 4k) 2023

COLLABORATIONS

List of large collaboration involvement working on major observational campaigns.

ARKS: ALMA Large Program to image debris disks at high resolution, P.I. Sebastian Marino

- *Role: Leading Radial Structure team. Our findings are available on arkslp.org.*

JWST High Contrast Imaging: JWST ERS program imaging exoplanets and disks, P.I. Sasha Hinkley

- *Role: Modeling MIRI 4-quadrant phase mask observations and VLT observations of HD 141569*

DustERS: JWST ERS program on Wolf-Rayet binary dust formation, P.I. Ryan Lau

- *Role: Developing spiral dust model for MIRI imaging published in Lau et al. 2022.*

COMMUNITY INVOLVEMENT

Referee & Peer Review

Referee for MNRAS and AAS Journals
ALMA distributed peer review, Cycles 8 – 12
ESO distributed peer review

Community & Outreach

Caltech Summer Undergraduate Research Fellowship Reviewer 2026
Conference chair, Cambridge Interdisciplinary Forum for Postgraduate Research, University of Cambridge 2024
Public talk speaker, Institute of Astronomy, Cambridge open evenings 2022 - 2023
Technology Officer, Gates Cambridge Scholars Council 2020 - 2021
General Executive, Sydney University Mathematics Society 2017 - 2019
Ambassador (co-organizing outreach events), Australian Science Innovations (nonprofit in education) 2016 - 2019
Interviewer, Berkeley Scientific Journal, UC Berkeley 2017

Other Employment

Supervisor (tutorial teaching), Institute of Astronomy, Cambridge 2022 - 2023
Supervisor (tutorial teaching), Trinity College, Cambridge 2020 - 2021
Tutor (tutorial teaching), School of Physics, University of Sydney 2019
Strategist, Goldman Sachs 2018

TALKS

Conference talks

- Exoplanets and Planet Formation *Dec 2025*
  The high-resolution structure of debris disks in the ARKS ALMA program. *Shanghai, China*

- Exploring Star–Planet–Disk Connections — *Oct 2025*
  High-resolution ALMA imaging of extrasolar planetesimal belts. — *Tartu, Estonia*
- Europlanet Science Congress 2024 — *Sep 2024*
  The high-resolution structure of debris disks revealed by the ARKS ALMA program. — *Berlin, Germany*
- Raising the veil on star formation near and far — *Apr 2024*
  High-resolution ALMA imaging of debris disks with the ARKS program. — *Cambridge, UK*
- Dust Devils: Debris Disks in the Sonoran Desert — *Mar 2024*
  The radial structure of debris disks in the ARKS ALMA program. — *Tucson, USA*
- Cosmic Dust 2023 — Aug 2023
  The physics of dusty Wolf-Rayet binaries revealed by observations of WR140. — *Kitakyushu, Japan*
- UK Exoplanet Community Meeting 2022 — Sep 2022
  Recovering the structure of edge-on debris disks non-parametrically. — *Edinburgh, UK*
- Debris Disks at Home and Abroad — Sep 2022
  Recovering the structure of edge-on debris disks non-parametrically. — *Jena, Germany*

Seminars

- MPIA Planet and Star Formation Seminar — *Oct 2025*
  The formation and evolution of dust in the colliding-wind binary Apep revealed by JWST. — *Heidelberg, Germany*
- Tsinghua Exoplanets and Planet Formation Seminar — *Sep 2025*
  Imaging debris disks with ALMA and JWST. — *Beijing, China*
- Columbia Astronomy & Astrophysics Seminar — *Mar 2025*
  Resolving radial substructures in debris disks. — *New York City, USA*
- JPL Astrophysics Luncheon Seminar — *Feb 2025*
  Interpreting the Resolved Structure of Debris Disks. — *Pasadena, USA*
- Department of Applied Mathematics and Theoretical Physics, University of Cambridge — *Oct 2023*
  The radial and vertical structure of debris disks. — *Cambridge, UK*
- Institute of Astronomy, University of Cambridge — Oct 2022
  Stellar fireworks: the spiral dust plumes of Wolf-Rayet binaries. — *Cambridge, UK*
- Institute of Astronomy, University of Cambridge — Mar 2022
  A non-parametric method to recover the structure of edge-on debris disks. — *Cambridge, UK*

Outreach talks

- Institute of Astronomy Open Evening — Oct 2023
  What are exoplanets doing? Asteroid belts give us a clue. — *Cambridge, UK*
- Institute of Astronomy Open Evening — Jan 2023
  Stellar fireworks: a journey through the spirally dust of massive binary stars. — *Cambridge, UK*
- Integrated Science Class Seminar (Year 6–8), Ming Cheng School — Apr 2022
  Planets of the Solar System and beyond. — *Beijing, China*
- Gates Cambridge Teach-a-thon — Feb 2021
  Planets of the Solar System and beyond. — *Online*

- Australian Council of Undergraduate Research — Oct 2019
  Highly anisotropic stellar winds in recently-identified Wolf-Rayet binary. — *Newcastle, Australia*

Poster presentations

- Gordon Research Conference on the Origins of Solar Systems — Jun 2025
  The radial structure of debris disks in the ARKS ALMA program. — *South Hadley, USA*
- Gordon Research Seminar + Conference on the Origins of Solar Systems — Jun 2023
  Modeling the structure of debris disks and the dust clump of Beta Pictoris. — *South Hadley, USA*
- STScI Spring Symposium: Planetary Systems and the Origins of Life in the Era of JWST — May 2023
  Modeling the structure of debris disks and the dust clump of Beta Pictoris. — *Baltimore, USA*
- Protostars and Planets VII — Apr 2023
  Has the dust clump in the debris disk of Beta Pictoris moved? — *Kyoto, Japan*

PUBLICATIONS

Publications available on ORCID 0000-0002-2106-0403, ADS Library and Google Scholar.

First-author Publications

9. [*In review*] **Y. Han**, M. C. Wyatt, K. Y. L. Su, *et al.* A radially broad collisional cascade in the debris disk of γ Oph revealed by JWST. *Submitted to ApJ* (2026). doi.org/10.48550/arXiv.2601.15285

8. [*In press*] **Y. Han**, M. C. Wyatt, M. Jankovic, *et al.* The multiwavelength vertical structure of the archetypal β Pictoris debris disk. *Submitted to A&A* (2026). doi.org/10.48550/arXiv.2603.03540

7. **Y. Han**, E. Mansell, J. Jennings, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) II. The Radial Structure of Debris Disks. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556450

6. **Y. Han**, R. M. T. White, J. R. Callingham, *et al.* The formation and evolution of dust in the colliding-wind Wolf-Rayet binary Apep observed by JWST. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-4357/ae12e5

5. **Y. Han**, M. C. Wyatt, S. Marino. Recovering the structure of debris disks non-parametrically from images. *Monthly Notices of the Royal Astronomical Society* (2025). doi.org/10.1093/mnras/staf282

4. **Y. Han**, M. C. Wyatt, W. R. F. Dent. Has the dust clump in the debris disk of Beta Pictoris moved? *Monthly Notices of the Royal Astronomical Society* (2023). doi.org/10.1093/mnras/stac3769

3. **Y. Han**, P. G. Tuthill, R. M. Lau, A. Soulain. Radiation-driven acceleration in the expanding WR140 dust shell. ***Nature*** (2022). doi.org/10.1038/s41586-022-05155-5

2. **Y. Han**, M. C. Wyatt, L. Matrà. RAVE: a non-parametric method for recovering the surface brightness and height profiles of edge-on debris disks. *Monthly Notices of the Royal Astronomical Society* (2022). doi.org/10.1093/mnras/stac373

1. **Y. Han**, P. G. Tuthill, A. Soulain, J. R. Callingham, P. M. Williams, P. A. Crowther, B. J. S. Pope, B. Marcote. The extreme colliding-wind system Apep: resolved imagery of the central binary and dust plume in the infrared. *Monthly Notices of the Royal Astronomical Society* (2020). doi.org/10.1093/mnras/staa2349

Co-authored Publications

** Indicates projects led by students who I advised*

24. * D. Wu, **Y. Han**, *et al.* Constraining properties of dust formed in Wolf-Rayet binary WR 112 using mid-infrared and millimeter observations. *The Astrophysical Journal* (2026). doi.org/10.3847/1538-4357/ae31f1

23. S. Marino, L. Matrà, A. M. Hughes, J. Ehrhardt, G. M. Kennedy, C. del Burgo, A. Brennan, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) I. Motivation, Sample, Data Reduction and Results Overview. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556489

22. B. Zawadzki, A. Fehr, A. M. Hughes, E. Mansell, J. Kittling, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) III. The Vertical Structure of Debris Discs. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556505

21. S. Mac Manamon, L. Matrà, S. Marino, A. Brennan, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) IV. CO gas imaging and overview. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556605

20. J. Milli, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) V. Comparison between scattered light and thermal emission. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556523

19. J. B. Lovell, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) VI. Asymmetries and Offsets. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556568

18. A. Brennan, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) VII. Optically Thick Gas with Broad CO Gaussian Local Line Profiles in the HD 121617 Disc. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556566

17. S. Marino, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) VIII. A dust arc and non-Keplerian gas kinematics in HD 121617. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556493

16. P. Weber, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) IX. Gas-driven origin for the continuum arc in the debris disc of HD121617. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556855

15. M. Jankovic, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) X. Interpreting the peculiar dust rings around HD 131835. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556637

14. R. M. T. White, B. J. S. Pope, P. G. Tuthill, **Y. Han**, *et al.* The Serpent Eating Its Own Tail: Dust Destruction in the Apep Colliding-Wind Nebula. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-4357/adfbe1

13. J. D. Monnier, **Y. Han**, *et al.* Revealing the Accelerating Wind in the Inner Region of Colliding-wind Binary WR 112. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-3881/adfa03

12. N. D. Richardson, *et al.* Carbon-rich dust injected into the interstellar medium by Galactic WC binaries survives for hundreds of years. *The Astrophysical Journal* (2025). 10.3847/1538-4357/addf30

11. M. Sommer, M. C. Wyatt, **Y. Han**. A PR drag origin for the Fomalhaut disk's pervasive inner dust: constraints on collisional strengths, icy composition, and embedded planets. *Monthly Notices of the Royal Astronomical Society* (2025). doi.org/10.1093/mnras/staf494

10. E. P. Lieb, *et al.* Dynamic imprints of colliding-wind dust formation from WR140. *The Astrophysical Journal Letters* (2025). doi.org/10.3847/2041-8213/ad9aa9

9. R. M. Lau, *et al.* A first look with JWST aperture masking interferometry (AMI): resolving circumstellar dust around the Wolf-Rayet binary WR 137 beyond the Rayleigh limit. *The Astrophysical Journal* (2024). doi.org/10.3847/2041-8213/ad9aa9

8. J. Terrill, S. Marino, R. A. Booth, **Y. Han**, J. Jennings, M. C. Wyatt. Deprojecting and constraining the vertical thickness of exoKuiper belts. *Monthly Notices of the Royal Astronomical Society* (2023). doi.org/10.1093/mnras/stad1847

7. R. M. Lau, *et al.* From dust to nanodust: resolving circumstellar dust from the colliding-wind binary Wolf-Rayet 140. *The Astrophysical Journal* (2022). doi.org/10.3847/1538-4357/acd4c5

6. R. M. Lau, M. J. Hankins, **Y. Han**, *et al.* Nested dust shells around the Wolf–Rayet binary WR 140 observed with JWST. *Nature Astronomy* (2022). doi.org/10.1038/s41550-022-01812-x

5. P. A. Robinson, X. Gao, **Y. Han**. Relationships between lognormal distributions of neural properties, activity, criticality, and connectivity. *Biological Cybernetics* (2021). doi.org/10.1007/s00422-021-00871-z

4. B. Marcote, J. R. Callingham, M. De Becker, P. G. Edwards, **Y. Han**, R. Schulz, J. Stevens, P. G. Tuthill. AU-scale radio imaging of the wind collision region in the brightest and most luminous non-thermal colliding wind binary Apep. *Monthly Notices of the Royal Astronomical Society* (2021). doi.org/10.1093/mnras/staa3863

3. R. M. Lau, M. J. Hankins, **Y. Han** *et al.* Resolving periodic spirals and shadows from the Wolf-Rayet dust factory WR112. *The Astrophysical Journal* (2020). doi.org/10.3847/1538-4357/abaab8

2. J. R. Callingham, P. A. Crowther, P. M. Williams, P. G. Tuthill, **Y. Han**, B. J. S. Pope. Two Wolf-Rayet stars at the heart of colliding-wind binary Apep. *Monthly Notices of the Royal Astronomical Society* (2020). doi.org/10.1093/mnras/staa1244

1. M. Li, **Y. Han**, M. J. Aburn, M. Breakspear, R. Q. Poldrack, J. M. Shine, J. T. Lizier. Transitions in information processing dynamics at the whole-brain network level are driven by alterations in neural gain. *PLOS Computational Biology* (2019). doi.org/10.1371/journal.pcbi.1006957